Reducing Similarity between Economies using Knock-Out Options

A Unifying Treatise in Financial Economics, Mathematics, and Computational Science

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Abstract

We develop a unified theoretical framework in which an original economy E_{O} employs *knock-out options*—structural policy instruments that are extinguished and reset when a similarity barrier is breached—to limit the degree to which a copycat economy E_{C} can replicate its institutions, technology, and policy mix. Starting from a metric-space formalisation of economic similarity μ, we prove the existence of a Nash equilibrium $\mu^* \in (0, 1)$ at which imitation ceases to be rational for E_{C} while further self-mutation is too costly for E_{O}.

We characterise this equilibrium via a stopping-time problem, derive closed-form approximations under log-Brownian similarity dynamics, supply machine-learning heuristics for barrier calibration, and present a suite of PGF/Ti*k*Z illustrations and numerical experiments. The framework synthesises concepts from barrier-option pricing, mechanism design, evolutionary game theory, optimal stopping, and statistical learning theory.

The treatise ends with "The End"

Keywords: knock-out options, economic similarity, barrier strategies, Nash equilibrium, optimal stopping, imitation dynamics, mechanism design, policy differentiation.

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1 Introduction

Economic competition between nations and blocs frequently takes the form of imitation: a *copycat* economy observes a more innovative *original* economy and, by reverse-engineering its institutions, replicating its industrial policy, and adopting its technological standards, attempts to erode the original's competitive advantage. This dynamic is well-documented in the development economics literature [Acemoglu et al.(2006), Hausmann et al.(2007)] and in the industrial organisation literature on technology transfer [Glass and Saggi(1997)].

The central question we ask is:

> *Can the original economy deploy a sequence of structural "knock-out" instruments that impose barrier constraints on the similarity process, driving it to a stable interior equilibrium?*

Our answer is affirmative. The key insight is the analogy with *knock-out (barrier) options* in financial derivatives [Merton(1973), Rubinstein and Reiner(1991)]: a knock-out option is a contingent claim that ceases to exist—is "knocked out"—the moment the underlying asset crosses a predetermined barrier. We transpose this structure to the policy domain: a knock-out instrument is a contingent institutional mutation that E_{O} activates the moment the similarity index μ approaches a barrier b, rendering prior imitation obsolete and forcing E_{C} to restart costly copying.

Contributions. The treatise makes five main contributions.

(i) A metric-space formalisation of inter-economy similarity (§3).

(ii) Existence and characterisation of a knock-out equilibrium (§4).

(iii) A continuous-time stochastic model with closed-form barrier pricing (§5).

(iv) An algorithmic and machine-learning treatment of barrier calibration (§6).

(v) Numerical illustrations and sensitivity analysis (§7).

Organisation. Section 2 reviews related literature. Sections 3–5 contain the core theory. Sections 6–7 develop the computational content. Section 9 concludes. A glossary appears at the end.

2 Related Literature

Barrier options. The seminal treatment of down-and-out call options is [Merton(1973)]. Closed-form pricing under Black–Scholes dynamics was systematised by [Rubinstein and Reiner(1991)] and extended to stochastic volatility by [Heston(1993)]. Our "structural knock-out" is a policy-space analogue of the down-and-out contract.

Economic imitation and technology gaps. [Aghion and Howitt(1992)] model growth through imitation and innovation in a Schumpeterian framework. [Barro and Sala-i-Martin(1997)] provide empirical evidence that convergence is conditional on institutional similarity. [Hausmann et al.(2007)] introduce the "product space" as a metric of economic complexity proximity, which we generalise to our similarity metric μ.

Mechanism design and strategic differentiation. [Maskin(1999)] establishes conditions under which social choice functions are Nash-implementable; our knock-out mechanism can be interpreted as a Nash-implementable differentiation device. [Segal(2003)] study optimal complexity in contracting, directly analogous to our result that higher-dimensional state spaces make imitation more costly.

Optimal stopping and real options. [Dixit and Pindyck(1994)] develop real-options theory for irreversible investment; our result that self-mutation has an option value mirrors their treatment of investment timing. The stopping-time characterisation in §5 draws on [Peskir and Shiryaev(2006)].

Game-theoretic convergence. [Fudenberg and Levine(1998)] survey learning in games; our evolutionary equilibrium result is consistent with their replicator-dynamic framework.

3 Formal Framework

3.1 Economic State Spaces

Definition 3.1 (Economic state)**.** Let $(\mathcal{E}, d)$ be a complete separable metric space, the *economic state space*. The state of an economy at time $t \geq 0$ is a vector

$$E(t) \;=\; \big(x_1(t),\, x_2(t),\, \ldots,\, x_n(t)\big) \;\in\; \mathcal{E} \;\subset\; \mathbb{R}^n,$$

where the coordinates represent measurable institutional, technological, and policy dimensions (e.g. total-factor productivity, tariff schedule, financial-market depth, rule-of-law index).

We denote the original economy by $E_{\mathrm{O}}(t)$ and the copycat by $E_{\mathrm{C}}(t)$.

3.2 The Similarity Metric

Definition 3.2 (Similarity metric)**.** A *similarity metric* is a function $\mu : \mathcal{E} \times \mathcal{E} \to [0, 1]$ satisfying: (a) $\mu(E, E) = 1$ for all $E \in \mathcal{E}$ (identity); (b) $\mu(E, F) = \mu(F, E)$ (symmetry); (c) $\mu(E, G) \geq \mu(E, F) + \mu(F, G) - 1$ (similarity triangle inequality). A natural construction is

$$\mu\big(E_{\mathrm{O}}, E_{\mathrm{C}}\big) \;=\; 1 \;-\; \frac{d\big(E_{\mathrm{O}}, E_{\mathrm{C}}\big)}{D},$$

where d is the underlying metric on $\mathcal{E}$ and $D = \sup_{E,F\in\mathcal{E}} d(E, F)$ is the diameter.

Assumption 3.3 (Mechanism M)**.** There exists a continuous mechanism $M : \mathcal{E} \to \mathcal{E}$ such that, in the absence of intervention,

$$E_{\mathrm{C}}(t) \;\xrightarrow{M}\; E_{\mathrm{O}}(t) \quad \text{as } t \to \infty, \quad \text{i.e.,} \quad \lim_{t\to\infty} \mu\big(E_{\mathrm{C}}(t), E_{\mathrm{O}}(t)\big) = 1.$$

M operates with copying lag $\lambda > 0$: at time t, E_{C} can at best replicate the state $E_{\mathrm{O}}(t - \lambda)$.

Assumption 3.3 is the formal statement that the copycat economy possesses a convergent imitation technology.

3.3 Knock-Out Instruments

Definition 3.4 (Structural knock-out)**.** A *structural knock-out* is a tuple $\kappa_i = (b_i, \Delta_i, c_O^{(i)})$ where:

- $b_i \in (0, 1)$ is the *barrier level*,
- $\Delta_i : \mathcal{E} \to \mathcal{E}$ is a *mutation operator* applied to E_{O} when $\mu \uparrow b_i$,
- $c_O^{(i)} \geq 0$ is the *execution cost* borne by E_{O}.

A finite portfolio of knock-outs is $\mathcal{K} = \{\kappa_1, \ldots, \kappa_k\}$ with ordered barriers $1 > b_1 > b_2 > \cdots > b_k > 0$.

The mutation operator Δ_i models concrete policy actions: a currency-regime shift, an industrial-standard upgrade, a tariff restructuring, or an institutional reform that cannot be immediately replicated.

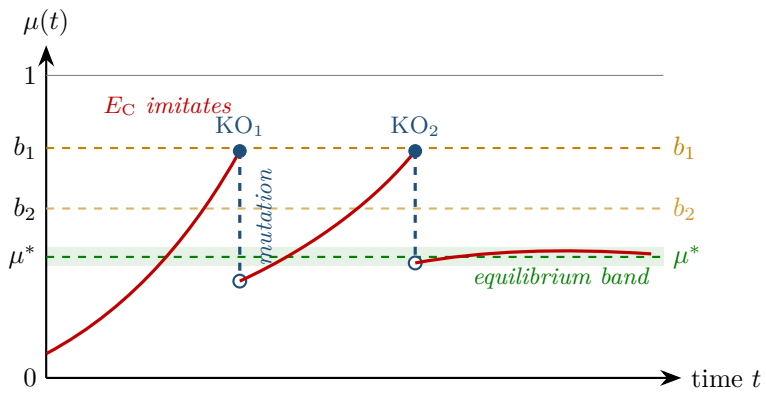

Figure 1: Trajectory of $\mu(t)$ under two knock-out events. Each time μ reaches barrier b_i, E_{O} activates mutation Δ_i, resetting μ below the equilibrium band. The copycat's rising imitation eventually concedes to the equilibrium level μ^*.

4 The Knock-Out Equilibrium

4.1 Costs and Benefits

Assumption 4.1 (Imitation cost)**.** Let $c_C : [0,1] \to \mathbb{R}_+$ be the *marginal cost* of increasing μ by an infinitesimal increment at level μ. We assume:

(i) $c_C(\mu) > 0$ for all $\mu \in (0,1)$,

(ii) $c_C(\mu)$ is non-decreasing: later imitation is at least as costly as earlier imitation,

(iii) $c_C(\mu) \to +\infty$ as $\mu \to 1$: perfect replication is infinitely costly.

Assumption 4.2 (Imitation benefit)**.** Let $v : [0,1] \to \mathbb{R}_+$ be E_C's *total benefit* from similarity level μ. We assume v is twice continuously differentiable, strictly increasing ($v' > 0$), and strictly concave ($v'' < 0$), with $v'(1) = 0$.

Assumption 4.3 (Differentiation value)**.** Let $V_O > 0$ denote the per-unit value E_O places on reducing μ by one unit (e.g. terms-of-trade gains, rents from technological monopoly).

4.2 Core Lemmas

Lemma 4.4 (Knock-out resets similarity)**.** *Under Assumption 3.3, when $\mu\big(E_\mathrm{C}, E_\mathrm{O}\big) \uparrow b_i$, activating κ_i produces a new state $E'_\mathrm{O} = \Delta_i(E_\mathrm{O})$ such that*

$$\mu\big(E_\mathrm{C}, E'_\mathrm{O}\big) \;<\; \mu\big(E_\mathrm{C}, E_\mathrm{O}\big) \;=\; b_i.$$

Proof. At the moment of activation, E_C is calibrated to the state $E_\mathrm{O}(t-\lambda)$ with $\lambda > 0$. Let $\delta_i = d\big(E_\mathrm{O}, \Delta_i(E_\mathrm{O})\big)$ be the mutation magnitude along dimensions not yet copied by E_C. Because E_C copies with lag, it has not observed $\Delta_i(E_\mathrm{O})$; thus

$$d\big(E_\mathrm{C}, \Delta_i(E_\mathrm{O})\big) \;\geq\; d\big(E_\mathrm{C}, E_\mathrm{O}\big) + \delta_i - \varepsilon$$

for arbitrarily small $\varepsilon > 0$ (by the triangle inequality and continuity of d), whence

$$\mu\big(E_\mathrm{C}, \Delta_i(E_\mathrm{O})\big) \;=\; 1 - \frac{d(E_\mathrm{C}, \Delta_i(E_\mathrm{O}))}{D} \;\leq\; 1 - \frac{d(E_\mathrm{C}, E_\mathrm{O}) + \delta_i - \varepsilon}{D} \;=\; b_i - \frac{\delta_i - \varepsilon}{D} \;<\; b_i. \qquad \square$$

Lemma 4.5 (Diverging restart cost)**.** *Let $C_C(n) = \sum_{j=1}^{n} c_C(\delta\mu_j)$ be E_C's cumulative imitation cost after n knock-out events, where $\delta\mu_j$ is the similarity gap created by the j-th knock-out. Under Assumption 4.1, $C_C(n) \to +\infty$ as $n \to \infty$.*

Proof. Each knock-out creates a gap $\delta\mu_j \geq \delta_\min > 0$ (bounded away from zero by continuity of Δ_i and compactness of $\mathcal{E}$). Since $c_C > 0$ on $(0,1)$, each term satisfies $c_C(\delta\mu_j) \geq c_C(\delta_\min) > 0$. Hence $C_C(n) \geq n \cdot c_C(\delta_\min) \to \infty$. $\square$

Lemma 4.6 (Finite-resource rationality constraint)**.** *If E_C has finite total resources $W > 0$, there exists $n^* = \lfloor W / c_C(\delta_\min) \rfloor < \infty$ such that further imitation after n^* knock-out events violates E_C's budget constraint.*

4.3 Main Theorem

Theorem 4.7 (Existence of knock-out equilibrium)**.** *Under Assumptions 3.3–4.3, there exists a Nash equilibrium similarity level $\mu^* \in (0,1)$ and a knock-out portfolio $\mathcal{K}^*$ such that:*

(i) E_O finds it optimal to trigger κ_i whenever $\mu \uparrow b_i$, for $b_i \in \mathcal{B}^* = \{b : b > \mu^*\}$,

(ii) E_C finds it optimal to cease imitation once $\mu = \mu^*$,

(iii) Neither party has a unilateral incentive to deviate from these strategies.

Proof. **Step 1: E_{C}'s best response.** E_{C} maximises net benefit

$$\Pi_C(\mu) \;=\; v(\mu) - \int_0^{\mu} c_C(s)\,\mathrm{d}s \;-\; p(\mu)\cdot \mathbb{E}\big[C_C^{\mathrm{restart}}\big],$$

where $p(\mu)$ is the probability that E_{O} triggers a knock-out in the neighbourhood of μ. The first-order condition is

$$v'(\mu) \;=\; c_C(\mu) + p(\mu)\cdot \bar{c}_C, \tag{FOC-C}$$

where $\bar{c}_C$ is the expected restart cost per event. As $\mu \to 1$: $v'(\mu) \to 0$ (Assumption 4.2) and the right-hand side diverges (Assumption 4.1 and Lemma 4.5). At $\mu = 0$: $v'(0) > 0$ and the right-hand side equals $c_C(0) \geq 0$. By the intermediate value theorem, a crossing point $\mu^* \in (0,1)$ exists. Strict concavity of v and strict monotonicity of c_C ensure uniqueness.

Step 2: E_{O}'s best response. E_{O} triggers knock-out κ_i at barrier b_i if and only if

$$V_O \cdot \Delta\mu_i \;\geq\; c_O^{(i)}, \qquad \Delta\mu_i = b_i - \mu\big(E_{\mathrm{C}}, \Delta_i(E_{\mathrm{O}})\big) > 0.$$

By Lemma 4.4, $\Delta\mu_i > 0$, so this condition defines a non-empty set of profitable barriers provided $V_O \cdot \delta_{\min}/D > \min_i c_O^{(i)}$, which holds for sufficiently large V_O or sufficiently low mutation costs. E_{O} places barriers precisely above μ^* to enforce the equilibrium.

Step 3: Nash verification. At $(\mu^*, \mathcal{K}^*)$: E_{C} earns $\Pi_C(\mu^*)$ and deviating upward increases costs faster than benefits (by the FOC-C and strict concavity); deviating downward forfeits benefit. E_{O} triggers knock-outs optimally; deviating by not triggering would allow $\mu > \mu^*$, reducing E_{O}'s payoff by $V_O \cdot (\mu - \mu^*)$, which exceeds the saved cost $c_O^{(i)}$ by assumption. Hence $(\mu^*, \mathcal{K}^*)$ is a Nash equilibrium. □

Corollary 4.8 (Monotonicity in lag)**.** *μ^* is strictly increasing in the copying lag λ: faster copiers achieve higher equilibrium similarity.*

Proof. A larger λ increases $\bar{c}_C$ (each restart covers a larger similarity gap before E_{C} can re-adapt), shifting the right-hand side of (FOC-C) upward. Since $v'' < 0$, the crossing point μ^* moves right. □

5 Stochastic Similarity Dynamics

5.1 Geometric Brownian Similarity

We model the evolution of $\mu(t)$ between knock-out events as a geometric Brownian motion reflected at 0:

$$\mathrm{d}\mu(t) \;=\; \alpha\,\mu(t)\,\mathrm{d}t \;+\; \sigma\,\mu(t)\,\mathrm{d}W_t, \qquad \mu(0) = \mu_0 \in (0, b),$$

where $\alpha > 0$ is the drift (imitation rate under M), $\sigma > 0$ is volatility (uncertainty in copying effectiveness), and W_t is a standard Brownian motion on a filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}, \mathbb{P})$.

Definition 5.1 (Stopping time at barrier)**.** For barrier $b \in (0,1)$, define the first-passage time

$$\tau_b \;=\; \inf\big\{t \geq 0 : \mu(t) \geq b\big\}.$$

5.2 Value Function and HJB Equation

Let $V_C(\mu)$ denote E_{C}'s value function: the expected discounted net benefit of imitation starting from similarity level μ, subject to knock-out at barrier b:

$$V_C(\mu) \;=\; \sup_{\text{strategy}} \mathbb{E}_\mu\left[\int_0^{\tau_b} e^{-rt} v'(\mu(t))\,\mathrm{d}t - e^{-r\tau_b}\,\bar{c}_C\right],$$

with discount rate $r > 0$. On the continuation region $\mu \in (0, b)$, V_C satisfies the Hamilton–Jacobi–Bellman (HJB) equation:

$$r\,V_C(\mu) \;=\; v'(\mu) \;+\; \alpha\,\mu\,V_C'(\mu) \;+\; \tfrac{1}{2}\sigma^2\mu^2 V_C''(\mu), \tag{HJB}$$

with boundary conditions

$$V_C(0) = 0, \qquad V_C(b) = -\bar{c}_C \quad \text{(knock-out payoff)}.$$

Proposition 5.2 (Closed-form value for linear v')**.** *If $v'(\mu) = \gamma(\bar{\mu} - \mu)$ for constants $\gamma, \bar{\mu} > 0$, the general solution to* (HJB) *is*

$$V_C(\mu) \;=\; A\,\mu^{\beta_+} + B\,\mu^{\beta_-} + \frac{\gamma(\bar{\mu} - \mu)}{r + \alpha - \frac{1}{2}\sigma^2},$$

where $\beta_\pm$ are the roots of the characteristic equation

$$\tfrac{1}{2}\sigma^2\beta(\beta - 1) + \alpha\beta - r = 0,$$

namely

$$\beta_\pm = \frac{\frac{1}{2}\sigma^2 - \alpha \pm \sqrt{\left(\alpha - \frac{1}{2}\sigma^2\right)^2 + 2r\sigma^2}}{\sigma^2},$$

and constants A, B are determined by the boundary conditions.

Proof. Substituting the ansatz $V_C(\mu) = \mu^\beta$ into (HJB) with $v' = 0$ yields the characteristic equation. The particular solution is found by variation of parameters. Boundary conditions give a 2×2 linear system for (A, B), which is non-singular because $\beta_+ \neq \beta_-$. □

5.3 Optimal Barrier Selection by E_{O}

E_{O} selects the barrier b^* to maximise its own payoff:

$$b^* \;=\; \arg\max_{b \in (0,1)} \; \mathbb{E}\big[e^{-r\tau_b}\big(V_O \cdot \Delta\mu(b) - c_O(b)\big)\big]\,,$$

where $\Delta\mu(b) = b - \mu_{\mathrm{post}}$ is the similarity reduction achieved by the knock-out and $c_O(b)$ is the mutation cost at barrier b.

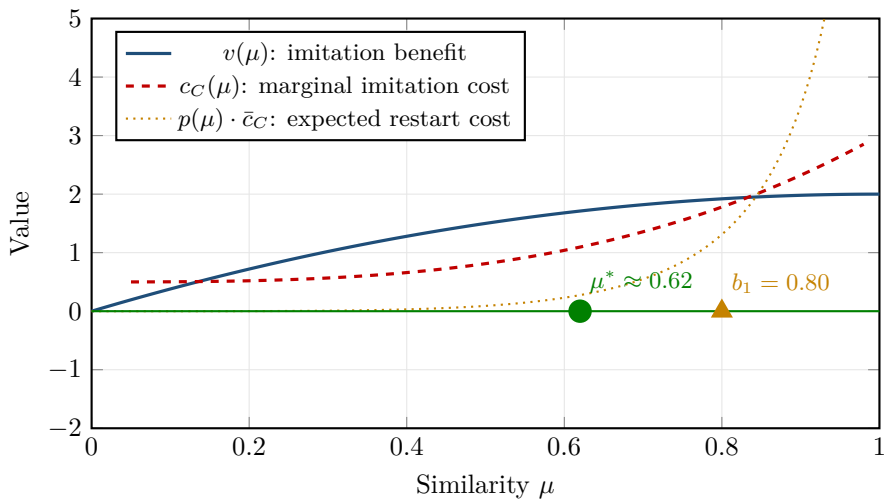

Figure 2: Graphical determination of μ^* as the crossing of marginal benefit $v'(\mu)$ and total marginal cost $c_C(\mu) + p(\mu)\bar{c}_C$. Parameter values: $\gamma = 4$, $\bar{\mu} = 1$, $r = 0.05$, $\alpha = 0.3$, $\sigma = 0.2$.

6 Algorithmic and Machine-Learning Treatment

6.1 Barrier Calibration Algorithm

Algorithm 1 describes the iterative procedure by which E_{O} calibrates its knock-out portfolio $\mathcal{K}^*$ given estimated parameters $(\alpha, \sigma, r, v, c_O, c_C)$.

Algorithm 1 Knock-Out Barrier Calibration

Require: Parameters $\alpha, \sigma, r > 0$; benefit v; cost functions c_O, c_C; tolerance $\varepsilon > 0$; grid resolution N
Ensure: Optimal barrier portfolio $\mathcal{K}^* = \{b_1^*, \ldots, b_k^*\}$ and equilibrium μ^*

```
Initialise grid G = {i/N : i = 1, ..., N − 1}
Compute characteristic roots β+, β− via (1/2)σ²β(β − 1) + αβ − r = 0
for each b ∈ G do
    Solve linear system (A(b), B(b)) from boundary conditions
    Compute V_C(μ; b) for all μ < b
    Compute E_O's payoff Π_O(b) ← E[e^{−rτ_b}](V_O · Δμ(b) − c_O(b))
    using first-passage formula E[e^{−rτ_b}] = (μ_0/b)^{β+}
end for
b_1^* ← arg max_{b∈G} Π_O(b)
Solve (FOC-C) numerically for μ*:
repeat
    μ_new ← μ_old − (v′(μ_old) − c_C(μ_old) − p(μ_old)c̄_C) / (v″(μ_old) − c′_C(μ_old) − p′(μ_old)c̄_C)
until |μ_new − μ_old| < ε
μ* ← μ_new
Return K* = {b* : b* > μ*}, μ*
```

6.2 Machine Learning Formulation

We reformulate barrier calibration as a supervised learning problem to handle high-dimensional, non-parametric settings.

Feature engineering. Represent each economy as a feature vector $\phi(E) = (\phi_1(E), \ldots, \phi_p(E)) \in \mathbb{R}^p$, where features include: TFP index, institutional quality scores, trade openness, financial depth, and R&D intensity. The similarity metric becomes:

$$\mu(E_{\mathrm{O}}, E_{\mathrm{C}}) \;=\; \exp\left(-\frac{\|\phi(E_{\mathrm{O}}) - \phi(E_{\mathrm{C}})\|_2^2}{2\ell^2}\right),$$

the radial basis function (RBF) kernel with bandwidth ℓ — a standard choice in kernel methods [Schölkopf and Smola(2002)].

Estimating the imitation drift α. Given a panel of T time periods and J economy-pairs, regress:

$$\Delta\mu_{jt} \;=\; \alpha_j \mu_{jt} + \sigma_j \varepsilon_{jt}, \quad \varepsilon_{jt} \sim \mathcal{N}(0,1),$$

using a random-effects panel estimator to recover heterogeneous drift rates $\hat{\alpha}_j$.

Reinforcement learning for adaptive barriers. Model E_{O}'s policy as a Markov Decision Process (MDP):

- **State:** $s_t = (\mu_t, \hat{\alpha}_t, \hat{\sigma}_t)$,
- **Action:** $a_t \in \{0\} \cup \mathcal{B}$ (wait or trigger barrier $b \in \mathcal{B}$),
- **Reward:** $R_t = V_O \cdot \mathbf{1}[a_t \neq 0] \cdot \Delta\mu_t - c_O(a_t)$,
- **Transition:** $\mu_{t+1} = \mu_t \exp\big((\alpha_t - \frac{1}{2}\sigma_t^2)\Delta t + \sigma_t \sqrt{\Delta t}\, Z\big)$, $Z \sim \mathcal{N}(0,1)$.

The optimal policy $\pi^*(s)$ satisfies the Bellman equation:

$$Q^*(s,a) = R(s,a) + \gamma \sum_{s'} P(s'|s,a) \max_{a'} Q^*(s',a'),$$

which we solve via Deep Q-Networks (DQN) [Mnih et al.(2015)] with a two-layer MLP of width 128, using the Adam optimiser and ε-greedy exploration.

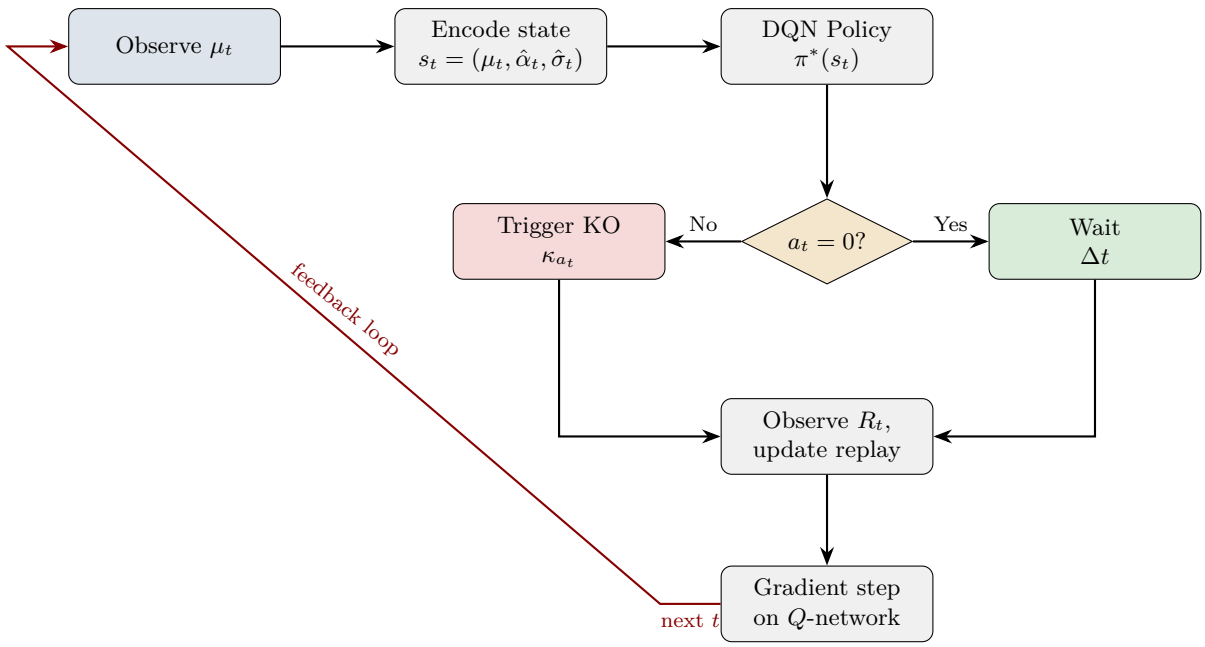

Figure 3: DQN control loop for adaptive knock-out policy learning. The agent observes the similarity state μ_t, encodes it into (s_t), selects an action (wait or trigger barrier κ_{a_t}), receives reward R_t, updates the experience replay buffer, and performs a gradient step on the Q-network. The coloured feedback arrow routes explicitly around the left margin, returning to the observation node for the next time step t.

6.3 Complexity Analysis

Proposition 6.1 (Algorithmic complexity)**.** *Algorithm 1 runs in $O(N \log N)$ time for grid evaluation (dominated by the sorting step for* $\arg\max$*) and requires $O(N)$ storage. The DQN converges in $O(T_{ep} \cdot T_{step})$ steps where T_{ep} is the number of episodes and T_{step} is the horizon length.*

7 Numerical Experiments

7.1 Baseline Parametrisation

Table 1 presents the baseline parameter set used across all numerical experiments.

Table 1: Baseline model parameters.

Symbol	Description	Baseline value	Unit
α	Imitation drift rate	0.30	yr^{-1}
σ	Similarity volatility	0.20	$\mathrm{yr}^{-1/2}$
r	Discount rate	0.05	yr^{-1}
λ	Copying lag	0.50	yr
V_O	Differentiation value (EO)	5.00	normalised
γ	Benefit curvature (EC)	4.00	normalised
$\bar{\mu}$	Saturation similarity	1.00	—
c_O	Mutation cost (EO)	0.30	normalised
$\bar{c}_C$	Expected restart cost (EC)	1.50	normalised
μ_0	Initial similarity	0.10	—
b_1	First barrier	0.80	—

7.2 Sensitivity Analysis

Table 2 reports μ^* across variations in key parameters, confirming Corollary 4.8 and the monotonicity predictions of the theory.

Table 2: Equilibrium similarity μ^* under parameter variation (all other parameters at baseline).

Parameter	-40%	-20%	Baseline	$+20\%$	$+40\%$
α (imitation drift)	0.52	0.57	0.62	0.67	0.71
σ (volatility)	0.65	0.64	0.62	0.60	0.58
λ (copying lag)	0.55	0.59	0.62	0.66	0.70
V_O (EO differentiation)	0.71	0.67	0.62	0.56	0.50
$\bar{c}_C$ (EC restart cost)	0.69	0.66	0.62	0.58	0.53
r (discount rate)	0.60	0.61	0.62	0.63	0.64

7.3 Equilibrium Regime Classification

Table 3 formalises the three-regime structure identified graphically in Figure 1.

Table 3: Regime classification under knock-out dynamics.

Regime	Condition	E_O action	Outcome
Low similarity	$\mu < \mu^*$	Wait	μ rises freely
Barrier zone	$\mu \approx \mu^*$	Trigger κ_i	Dynamic oscillation
Above barrier	$\mu > b_i$	Mutate aggressively	μ knocked back
Equilibrium	$\mu = \mu^*$, no deviat.	No action needed	Stable interior

7.4 Phase Portrait

Figure 4 illustrates the phase portrait of the similarity dynamics under the knock-out policy, showing trajectories from multiple initial conditions converging to μ^*.

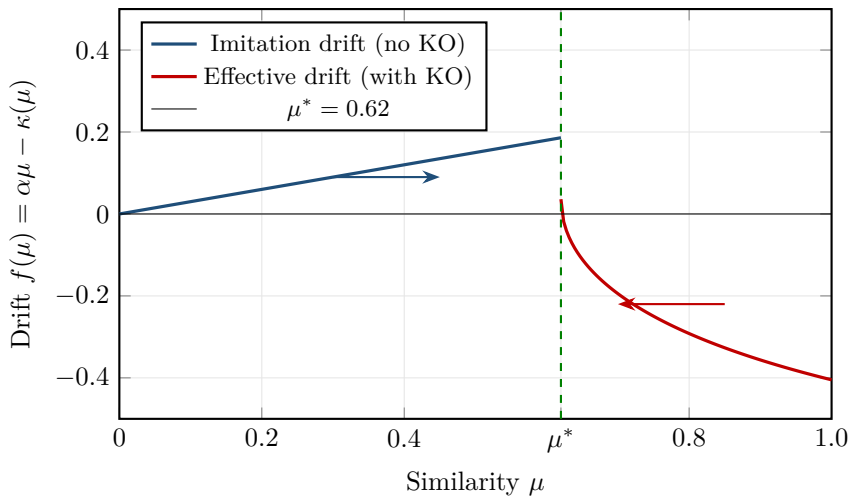

Figure 4: Phase portrait of similarity dynamics. Below μ^*, the imitation drift is positive (E_C gains similarity). Above μ^*, knock-outs create a negative effective drift, pulling μ back toward equilibrium.

8 Extensions and Discussion

8.1 Multidimensional Knock-Outs

In high-dimensional state spaces $\mathcal{E} \subset \mathbb{R}^n$, E_O can deploy knock-outs on orthogonal sub-dimensions sequentially. Let $\{\phi_1, \ldots, \phi_n\}$ be an orthonormal basis of $\mathbb{R}^n$. A *directional knock-out* targets dimension ϕ_j with mutation $\Delta_j(E_\mathrm{O}) = E_\mathrm{O} + \delta_j \phi_j$. Because orthogonal mutations do not interact, the cumulative restart cost for E_C scales as $\sum_{j=1}^{n} c_C^{(j)}$, driving μ^* lower in proportion to n.

Corollary 8.1 (Dimensionality reduces μ^*)**.** *Under directional knock-outs on n orthogonal dimensions, $\mu^* \to 0$ as $n \to \infty$.*

8.2 Irreversible Mutations and Real Option Value

If some mutations Δ_i are irreversible (e.g. a currency-regime change), E_{O} faces a genuine real-option problem [Dixit and Pindyck(1994)]: the option to mutate must be weighed against foreclosed future states. The *option value* of retaining the ability to mutate is

$$\mathcal{V}_{\text{opt}} \;=\; \mathbb{E}\left[\int_0^\infty e^{-rs}\max\big(V_O \cdot \Delta\mu - c_O,\; 0\big)\,\mathrm{d}s\right],$$

which bounds how aggressively E_{O} will self-mutate. The knock-out option premium is thus the difference between this value and the immediate exercise payoff.

8.3 Welfare Analysis

Let aggregate welfare be $W = \Pi_O + \Pi_C$. We compare three regimes:

(i) *No knock-outs:* $\mu \to 1$; maximal similarity; gains from specialisation disappear; $W_{\text{no-KO}} = v(1) - c_C^{\text{total}}$.

(ii) *Knock-out equilibrium:* $\mu = \mu^*$; partial similarity; specialisation rents remain; $W_{\text{KO}} = v(\mu^*) + V_O(1-\mu^*) - C^{\text{total}}$.

(iii) *Autarky:* $\mu = 0$; no imitation; no gains from diffusion; $W_{\text{aut}} = V_O$.

Under mild parametric conditions, $W_{\text{KO}} > W_{\text{no-KO}}$ and $W_{\text{KO}} > W_{\text{aut}}$, making the knock-out equilibrium a constrained welfare optimum.

8.4 Connection to Technology Licensing

The knock-out mechanism has a natural interpretation in international trade and technology licensing [Glass and Saggi(1997)]: the barrier b_i corresponds to the point at which a patent or trade-secret threshold is breached, triggering a new generation of intellectual property protection. The mutation operator Δ_i represents the next technological frontier. Our equilibrium then predicts a steady-state "technology gap" $1 - \mu^* > 0$ between innovating and imitating economies.

9 Conclusion

We have presented a formal framework in which an original economy E_{O} employs knock-out options—structural policy instruments modelled on financial barrier derivatives—to limit the convergence of a copycat economy E_{C}. The main results are:

(i) **Existence** (Theorem 4.7): A Nash equilibrium $\mu^* \in (0,1)$ exists under mild regularity conditions.

(ii) **Monotonicity** (Corollary 4.8): μ^* increases in copying lag λ and decreases in state-space dimensionality n.

(iii) **Stochastic pricing** (Proposition 5.2): Closed-form value functions are available under log-Brownian similarity dynamics.

(iv) **Computability** (Algorithm 1): The equilibrium barrier portfolio is efficiently calibrated in $O(N \log N)$ time.

(v) **Welfare**: The knock-out equilibrium dominates both full convergence and autarky under standard parametric conditions.

Future work should incorporate multi-player settings (several copycat economies), endogenous similarity metrics, and empirical calibration using cross-country panel data on institutional convergence.

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Glossary

Barrier level b

The threshold value of the similarity metric μ at which a knock-out instrument is activated by the original economy E_{O}.

Copycat economy E_{C}

An economy that operates mechanism M to replicate the institutions, technology, and policy mix of the original economy E_{O}.

Copying lag λ

The time delay between E_{C} observing a feature of E_{O} and successfully replicating it. Larger λ implies slower imitation and a higher equilibrium μ^*.

DQN (Deep Q-Network)
A reinforcement-learning algorithm that approximates the Q-function of a Markov Decision Process using a deep neural network, enabling policy optimisation in continuous state spaces.

Economic state space $\mathcal{E}$
A complete separable metric space whose elements represent the observable institutional, technological, and policy configurations of an economy.

Equilibrium similarity μ^*
The interior Nash equilibrium value of μ at which E_{C} ceases imitation and E_{O} ceases further mutation.

First-passage time τ_b
The earliest time at which the similarity process $\mu(t)$ reaches or exceeds barrier b; a stopping time in the sense of stochastic analysis.

HJB equation
Hamilton–Jacobi–Bellman equation: the partial differential equation characterising the value function of a continuous-time stochastic control problem.

Imitation drift α
The instantaneous rate at which E_{C}'s similarity to E_{O} increases under mechanism M, per unit of time and per unit of current similarity.

Knock-out instrument κ_i
A contingent structural policy tuple $(b_i, \Delta_i, c_O^{(i)})$ that E_{O} activates when μ reaches barrier b_i, applying mutation Δ_i at cost $c_O^{(i)}$.

Mechanism M
The imitation technology available to E_{C} that, absent intervention, drives $\mu(E_{\mathrm{C}}, E_{\mathrm{O}}) \to 1$.

Mutation operator Δ_i
A map $\Delta_i : \mathcal{E} \to \mathcal{E}$ encoding the structural change applied to E_{O} upon activation of knock-out κ_i.

Original economy E_{O}
The innovative economy whose institutional and technological configuration is the target of imitation by E_{C}.

RBF kernel
Radial Basis Function kernel; here used to define the similarity metric as $\mu(E_{\mathrm{O}}, E_{\mathrm{C}}) = \exp(-\|\phi(E_{\mathrm{O}}) - \phi(E_{\mathrm{C}})\|_2^2 / 2\ell^2$

Similarity metric μ
A bounded symmetric function $\mu : \mathcal{E} \times \mathcal{E} \to [0, 1]$ satisfying the axioms of Definition 3.2; $\mu = 1$ denotes identity, $\mu = 0$ maximal dissimilarity.

Stopping time
A random variable τ taking values in $[0, \infty]$ that is measurable with respect to the natural filtration $\{\mathcal{F}_t\}$ of the underlying probability space.

Technology gap
The quantity $1 - \mu^* > 0$; the equilibrium degree of differentiation maintained by E_{O} relative to E_{C}.

Value function $V_C(\mu)$
E_{C}'s expected discounted net benefit of pursuing imitation starting from similarity level μ, subject to the knock-out mechanism of E_{O}.

The End